

Name: _____

Date: _____

LEARNING INTENTION	examine why scientific advances often result from collaboration or build on earlier work
KEY IDEA	Science grows through shared observations, specialised roles, critique, replication and improved tools. Credit evidence and contributions rather than presenting discovery as one person's isolated moment.
WORKED MODEL	A field team records declining frog calls; laboratory scientists test water; Traditional Owners contribute long-term ecological knowledge; engineers improve sensors. Together, the evidence becomes stronger.

SUCCESS CRITERIA - I CAN

[] identify contributions [] trace how ideas build [] explain collaboration benefits and limits

A. NOTICE AND IDENTIFY

1. Match role to contribution: field observer, laboratory analyst, data specialist, community knowledge-holder.

2. Explain why repeating another team's test strengthens or challenges a finding.

3. Distinguish collaboration from merely dividing tasks.

B. APPLY AND EXPLAIN

4. Sequence: observation, shared question, different tests, comparison, revised explanation.

5. A report credits only the team leader. Identify missing evidence about contributions.

6. Explain how disagreement can improve science when claims are tested respectfully.

C. INDEPENDENT FLUENCY - CHECK AND CREATE

7. Create a contribution map for a habitat investigation with four roles and information exchanges.

8. Check whether the final conclusion depends on more than one type of expertise.

Aussie Primary Academy | Australian Curriculum v9.0 | Independent Application and Self-Review

Name: _____

Date: _____

D. APPLY THE SKILL IN CONTEXT

9. Case: two schools record heat data, a council supplies maps and an ecologist interprets tree cover. Explain each contribution.

10. Identify one reliability gain and one coordination challenge.

11. Write a reasoned claim explaining why the advance could not be credited to one source alone.

E. REASON, EVALUATE AND TRANSFER

12. Evaluate the claim 'Science advances because one genius finds the answer.' Use the case as evidence.

13. Design a collaboration plan that includes roles, shared records, review and fair acknowledgement.

F. STUDENT REVIEW - PROVE YOUR UNDERSTANDING

[] I can explain how collaboration and earlier work support scientific advances.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult decision was _____ because _____

One correction or improvement I made: _____

My next practice step: _____

Aussie Primary Academy | Australian Curriculum v9.0 | Teacher Copy - Answers and Teaching Support

Accept equivalent responses when the student applies the targeted skill accurately and justifies the choice.

COMPLETE ANSWER KEY

1. Roles contribute observations, controlled analysis, pattern interpretation and contextual/long-term knowledge.
2. Replication tests whether results are consistent rather than accidental.
3. Collaboration includes sharing, questioning, combining and revising ideas.
4. observation -> question -> tests -> comparison -> revised explanation.
5. Methods, data, interpretation and community knowledge contributions must be acknowledged.
6. Evidence-based disagreement exposes assumptions and prompts stronger tests.
- 7-8. Map must show information moving between roles and combined evidence.
9. Schools=data; council=spatial context; ecologist=relationship between heat and cover.
10. Gain: broader/triangulated evidence; challenge: consistent methods or communication.
- 11-13. Require evidence, balanced evaluation and explicit acknowledgement process.

TEACHER CHECK AND NEXT STEP

[] Uses accurate science ideas and evidence.

[] Explains a decision rather than listing an answer.

Next step: model one difficult item, complete one together, then transfer the skill to a new investigation.

COMMON MISCONCEPTION / WATCH OUT

Collaboration is not evidence that every claim is correct; methods still require testing.

Avoid the lone-genius story when advances clearly build on many contributions.

EXTENSION AND RE-TEACH

Re-teach: model one item aloud, complete one together, then ask the student to explain an independent correction.

Extension: transfer the skill to a short authentic text and justify two deliberate language choices with quoted evidence.